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AN APPLICATION OF PALÆOMAGNETISM IN ESTIMATING THE AGE OF ROCKS

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DESPITE the enormous amount of palæomagnetic data published over the past decade, there still exist considerable gaps on the polar wandering curves, which need to be re-adjusted before palæomagnetism can be claimed as a proved method of age determination of rocks. Nevertheless, even at this early stage, attempts to estimate the age of Palæozoic or younger rocks should not be underestimated, mainly in view of the striking coherency that characterizes most of the palæomagnetic measurements carried out so far.

The data presented in this article were derived from a suite of 20 oriented samples that were collected from undisturbed diabasic rocks from Ellef Ringnes Island at latitude 78° 45′ N. and longitude 103° 45′ W. (Fig. 1). An estimate of the time of intrusion of these rocks was obtained from a comparison of the pole position derived from their palæomagnetism with other dated pole positions for the North American continent.

The diabase rocks are known[1] to intrude Palæozoic and Mesozoic formations throughout the Canadian Arctic Archipelago. In the immediate vicinity of the sampling area, they intrude the Deer Bay Formation (Lowermost Cretaceous) as dykes and sills, whereas they cover the lower part of the Isachsen Formation (Lower Cretaceous) as sheets[2]. The upper age-limit of the diabase is not clearly defined, however, since there is no direct evidence indicating their relationship with respect to Tertiary sediments, although they are unconformably overlain by unconsolidated Pleistocene deposits. Up to now, the probable age of the diabase has been estimated as Lower or Upper Cretaceous on the basis that they have not been observed to intrude any of the Tertiary beds[1].

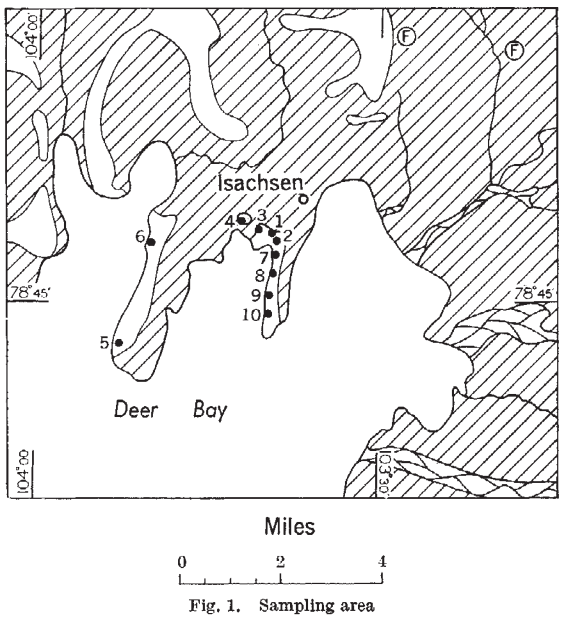


Fig. 1. Sampling area

F, fossil locality; ●, sampling sites. Hatched, undisturbed Cretaceous sediments; white, diabase and gabbro

Table 1. MEAN MAGNETIZATION DIRECTIONS FOR COLLECTING SITES

Collecting site	*N*	*S*	Initial magnetization directions *D*	*I*	*R*	Magnetization directions after alternating field treatment *D*	*I*	*R*
1	2	4	292·0	+72·0	3.99	290·0	+68·5	3·99
2	2	4	293·5	+75·5	4·00	246·5	+77·5	3·94
3	2	4	094·0	+77·5	3·80	110·5	+83·0	3·73
4	2	4	291·5	+83·0	3·99	253·0	+88·0	3·92
5	2	4	101·5	+86·0	3·92	192·5	+83·0	3·91
6	2	4	027·5	+85·0	3·98	052·0	+83·0	3·88
7	2	4	310·5	+77·0	3·92	290·0	+82·0	3·96
8	2	4	298·5	+57·0	3·98	292·0	+52·0	3·97
9	2	4	306·0	+71·5	4·00	302·5	+70·0	4·00
10	2	4	304·5	+74·5	3·90	265·0	+72·5	3·80

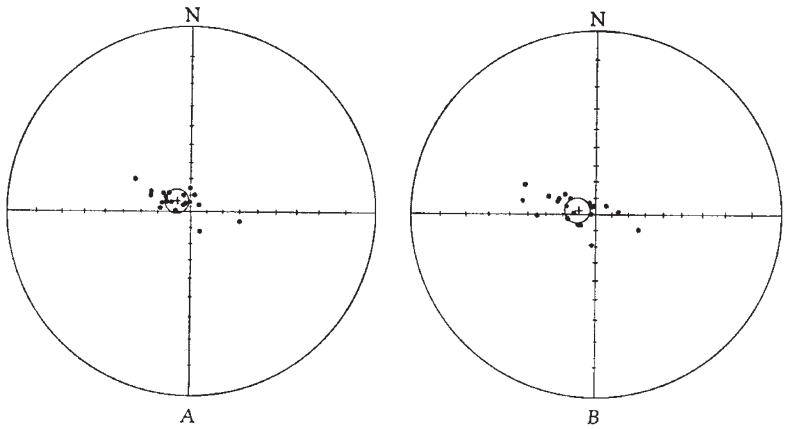
N, number of samples collected at each site; *S*, number of specimens from each site; *D*, mean site declination of magnetization; *I*, mean site inclination of magnetization; *R*, resultant of four magnetization vectors at each site.

For the collection, samples were oriented *in situ* with the aid of a solar compass and clinometer. At each of the ten sites visited (location shown in Fig. 1), two samples of unweathered rock were collected. Two 1-in. cubical specimens were cut out from each sample, two faces of each cube being parallel to the orientation mark and perpendicular to the collecting surface.

The natural remanent magnetization of each cube was determined both in intensity and direction by means of a spinner-type magnetometer recently developed at the Geological Survey of Canada.

The magnetic stability of the rocks was tested in an alternating field of 300 œrsteds (peak value). The change assumed by the mean directions of magnetization for each site as a result of this treatment was relatively insignificant, as may be verified from a comparison of columns 4 and 5 and columns 7 and 8 of Table 1. The mean directions of magnetization for each of the 20 samples before and after the alternating field treatment are plotted in Figs. 2*A* and *B* respectively. In the second of these groups of measurements the intensities of magnetization were lower than in the first group by about 10 per cent. It was concluded from these facts that the natural remanent magnetization of the rocks dealt with is predominantly of thermoremanent nature. On this basis,

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	D	I	N	∝	K	R
BEFORE TREATMENT (A)	306.0	+79.5	20	6.5	28	19.3
AFTER TREATMENT (B)	283.5	+79.0	20	7.5	20	19.1

Fig. 2. Stereographic projections of the directions of magnetization and circles of confidence before (*A*) and after (*B*) alternating field demagnetization
●, Vectors pointing downward; +, mean direction of magnetization

the mean direction of magnetization of the whole suite after the alternating field treatment was considered as indicative of the Earth's field attitude at the time the rocks were cooled from magmatic to normal surface temperatures.

With the assumption of a geocentric dipole field, the collecting site co-latitude was computed and the ancient pole position was derived from these data in terms of present geographic co-ordinates. The results of these computations are listed in Table 2 along with the half-angle of the cone of confidence at the 95 per cent probability-level[3].

Among the published pole positions derived from North American rocks, there appears to be a sharp break in the location of Tertiary with respect to Cretaceous poles. As may be verified from Fig. 3 and Table 2, the Tertiary poles are generally located at latitudes 80°–87° N., whereas the two Cretaceous poles available are located at latitudes in the vicinity of 70° N. One of the two North American Cretaceous poles is based on Runcorn's

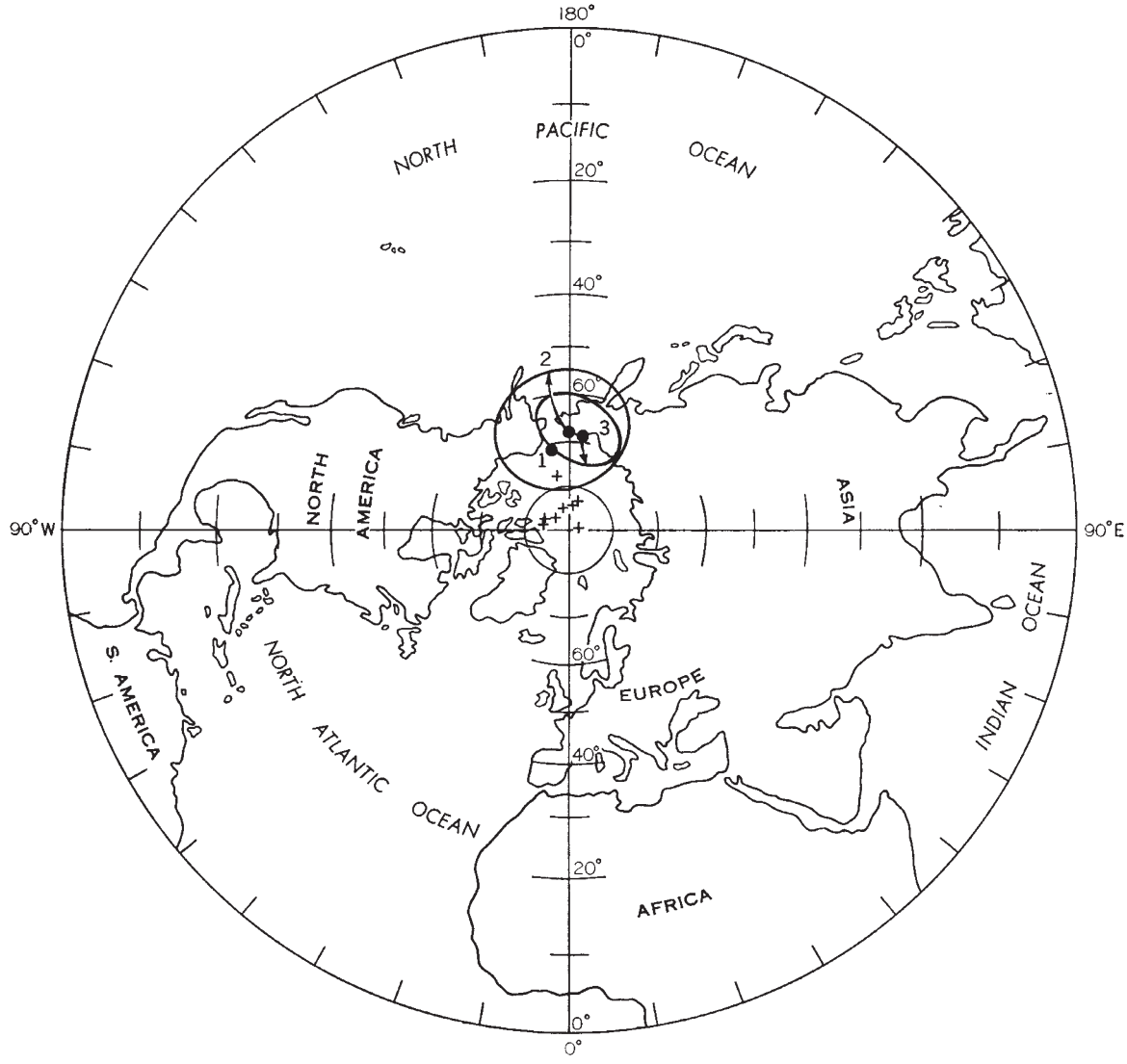


Fig. 3. Polar stereographic projection of Northern Hemisphere showing Cretaceous and Tertiary (+) pole positions and related ovals of confidence for poles 2 and 3. 1, Dakota sandstone; 2, Isachsen diabase; 3, Mount Megantic basic intrusives

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Table 2. Summary of Palæomagnetic Data for Cretaceous and Tertiary Rocks of North America

Formation	Age	Collection site Lat.	Long.	D	I	N	S	α_{95}	K	Pole position Lat.	Long.	$d\chi$	$d\psi$
Dakota sandstone †	Cretaceous	34 N.	110 W.	159·0	−60·5	3	9	—	—	72·5 N.	169·5 W.	—	—
Isachsen diabase	Cretaceous ?	79 N.	104 W.	283·5	+79·5	20	40	7·5	20	69 N.	180 W.	14·1	13·6
Basic intrusives Mount Megantic	Cretaceous	45·5 N.	71 W.	157·0	−52·0	12	24	6·5	44	68·5 N.	171·5 E.	8·9	6·1
Canadian basalts *	Late Tertiary	61 N.	134·5 W.	348·5	+75	46	—	4	28	85 N.	145 E.	7·2	6·7
Payette *	Miocene Plocene (?)	43 N.	115·5 W.	1·5	+62·5	13	—	2·5	258	88·5 N.	60 W.	4	3
Columbia River basalts *	Miocene	46·5 N.	120 W.	3·5	+69·5	73	—	8	4	83 N.	105 W.	13·5	11·5
Ellensburg *	Late Miocene Early Pliocene	46·5 N.	120·5 W.	0·5	+68·5	23	—	9	12	85 N.	115 W.	15·5	13
Duchesne River *	Tertiary	40·5 N.	110 W.	2	+65	24	85	5	14	83 N.	99 W.	8	6
Laney Shale *	Eocene	41·5 N.	109·5 W.	353·5	+62·5	19	—	6	30	85 N.	170 W.	9·5	7·5
Green River *	Eocene	39·5 N.	108 W.	345·5	+65·0	7	—	4·5	168	77·5 N.	158 W.	7	6
Wasatch *	Eocene	44·5 N.	109 W.	351·5	+63·5	4	—	17	30	84·5 N.	180 E.	26·5	21

N, number of samples; S, number of specimens; α, radius of the circle of confidence at the 95 per cent level of probability; K, estimate of the precision parameter; $d\chi$, $d\psi$, semi-axes of the oval of confidence; *, as reported by Cox and Doell (ref. 7); †, as calculated by us from the means of the three reversely polarized samples reported by Runcorn (ref. 4).

measurements of the Dakota Sandstone[4]. Runcorn considered that only the three reversely polarized samples of the six measured by him were magnetically stable. The mean magnetization direction was recomputed by us for these three samples, the mean of the first sample being based on four specimens, that of the second one on three specimens, and that of the third on two specimens. The recomputed mean direction for this group and the corresponding pole position are given in Table 2. The second Cretaceous pole available in the literature is based on the magnetization measurements of 12 samples[5] from the magnetically stable basic intrusive rocks of Mount Megantic. The age of these rocks has been established as Cretaceous by potassium/argon age determination[6].

The proximity of the poles derived from the Isachsen diabase and the Mount Megantic rocks, the magnetic stability of the rocks in both cases, and the general departure of these two poles and of the pole derived from the Dakota Sandstone from the Tertiary poles all indicate that the Isachsen diabase rocks are probably of Cretaceous age. Furthermore, this interpretation is not in contradiction with the date postulated by the available geological field evidence.

We thank the Director of the Geological Survey of Canada for allowing publication of this article and Mr. J. Babbin, of the Polar Continental Shelf Project, for collecting the Isachsen diabase samples.

Note added in proof. Since the manuscript for this article was first submitted, potassium/argon (whole rock) analyses were made for samples from sites 1 and 10. The ages obtained were 241 and 249 m.y. respectively, which would set the age of the dykes as Permian. This indication is fully incompatible with the biostratigraphic and palæomagnetic data (100 ± 25 m.y.) and no reasonable explanation has been found so far for this incompatibility.

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DECOMPOSITION OF SINGLE CRYSTALS OF LEAD HYDROXIDE IN THE ELECTRON MICROSCOPE

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In recent years there have been a number of investigations into the way in which solids decompose in the highly ionizing beam of the electron microscope. Compounds which have been examined include silver azide[1], magnesium hydroxide[2], silver iodide[3], lead iodide[4] and silver cyanamide[5]. This article describes some observations on the behaviour of lead hydroxide. This compound decomposes in an interesting manner with the formation of both crystallographic cavities and crystallographic nuclei of decomposition product.

The investigation arose out of a more general investigation of the structure and stability of lead and other inorganic cyanamides[6]. During the preparation of single crystals of lead cyanamide for examination in the electron microscope crystals of lead hydroxide are also formed. The latter are particularly suited for examination by transmission electron microscopy and by electron diffraction.

Preparation of Crystals

The preparation of crystals involved slow precipitation in alkaline solution. This was achieved by allowing ammonia gas to diffuse through a layer of ether covering a solution of lead cyanamide in dilute acetic acid. Before adding the ether, an evaporated carbon film was floated on to the surface of the acid solution and gold electron microscope grids (500 mesh) were placed in the solution on platinum gauze supports. Ether was then added to a depth of about 5 mm. The carbon film remained at the interface, though some of its coherence was lost. The vessel and its contents were then placed in a large desiccator containing a beaker of 880 ammonia. After 15–30 min a yellow precipitate of lead cyanamide was clearly visible and the vessel was then removed. Some crystals (not visible) are formed on the carbon film and this was then caught up over the grids by raising the support gauze through the film. Most of the liquid adhering to the grids was removed with filter paper, and, after drying, the grids were ready for examination in the microscope. This was a Siemens-Elmiskop I electron microscope operated at 80 kV.

Crystals of both lead cyanamide and lead hydroxide are to be found on the carbon support film. The former were identified by their electron diffraction pattern[7].

The crystals of lead hydroxide were clearly distinguishable both by their appearance and by their hexagonal diffraction patterns, which showed them to be in the (0001) orientation[8]. They were generally sufficiently thin for extinction contours to be visible and were hexagonal in shape (though the hexagons were seldom complete). Because of a strong tendency towards overgrowth, Moiré patterns were frequently observed, but dislocations were rarely seen. The possibility that the crystals were a basic lead salt seems unlikely as crystals with the same

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